

Rigetti Computing Inc

Superconducting Qubit Pioneer — Full-Stack Quantum Computing + Cloud Services

PhD Due Diligence Report | BLRI-DD-RGTI-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,453ch

PART I — SELDON THESIS AXIS

Superconducting Quantum: Speed vs Fidelity Trade-off

Rigetti occupies the highest-risk, highest-potential-upside position in the quantum computing landscape: a full-stack superconducting quantum company that builds its own chips, operates its own cloud (Quantum Cloud Services, QCS), and develops quantum software. This vertical integration is both its moat and its burn. In Seldon K-variable terms, Rigetti's superconducting approach — if it achieves fault tolerance — represents the most scalable path to general-purpose quantum computing that multiplies Knowledge Capital comprehensively.

Rigetti's qubit count (80+ qubits on Aspen-M-3) lags IBM (400+) and Google (Willow) in raw scale, but its full-stack ownership of chip fabrication (Fab-1 in Fremont, CA) is strategically unique among publicly traded quantum hardware companies. No other listed quantum hardware firm manufactures its own superconducting qubits in-house.

Source: moomoo OpenD [2026-08-13] · Rigetti investor relations · QCS platform

PART II — QUALITATIVE ANALYSIS

Fab-1: The Proprietary Superconducting Foundry

Rigetti's Fab-1 in Fremont, California is a dedicated 10,000 sq ft superconducting qubit fabrication facility — the only such facility owned by a publicly traded quantum company outside IBM or Google. This

means Rigetti controls its own process node, qubit geometry, and material science pipeline. The trade-off: enormous fixed cost structure. The benefit: no dependency on TSMC or external foundries for quantum chip production, and full IP ownership of process improvements.

Quantum Cloud Services (QCS): Developer Ecosystem

QCS provides direct access to Rigetti systems for developers, researchers, and enterprise customers. Its Pyquil language and Forest SDK are open-source and have accumulated a developer community. AWS Braket also hosts Rigetti hardware, providing market reach beyond direct QCS access. The developer ecosystem is an undervalued moat — switching costs from QCS increase as applications mature.

DARPA Quantum Benchmarking Initiative Contract

Rigetti won a DARPA contract under the Quantum Benchmarking Initiative to assess the path to quantum utility. This government validation is significant: DARPA funds only companies with credible technology trajectories. It also provides classified feedback on hardware performance that shapes Rigetti's roadmap.

Source: Rigetti 10-K EDGAR · DARPA programme announcements · AWS Braket partner page

PART III — QUANTITATIVE ANALYSIS

VALUATION SNAPSHOT [moomoo OpenD 2026-08-13]

Price: \$18.81 | Prev Close: \$18.42 | Mkt Cap: \$6.278B
PE TTM: N/M (loss) | EPS TTM: -\$0.70 | Net Income: -\$233.6M
Div Yield: 0% | 52w: \$12.53–\$58.15 | Shares: 333.8M
Historical close 2026-08-07: \$17.94 [BL3]

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PART IV — COMPETITIVE LANDSCAPE

Competitor	Modality	Mkt Cap	Rigetti Vs
IBM (not listed)	Superconducting	N/A (div)	IBM has 5× qubit count; Rigetti has Fab-1 independence
Google (not listed)	Superconducting	N/A (div)	Google Willow outperforms; Rigetti more accessible
IonQ (IONQ)	Trapped-ion	\$17.751B	IonQ has higher fidelity;

			Rigetti faster gates
D-Wave (QBSTS)	Annealing	\$7.873B	Different market — not direct competitor
Quantinuum (QNT)	Trapped-ion	\$2.561B	Both B2B; Quantinuum has Honeywell heritage

PART V — ETHNOGRAPHIC PORTRAITS

FINDING F-5-A: Portrait 1: The Quantum Algorithms Researcher (COMPOSITE)

A PhD researcher at Carnegie Mellon University uses Rigetti's Aspen-M system via QCS for variational quantum eigensolvers in materials science. He values QCS's low-latency direct access (vs AWS Braket's queue times) and Pyquil's flexibility for custom pulse-level control. His frustration: 'Rigetti's qubit count limits my circuit depth. IBM's 127-qubit Eagle is where I do serious work.' He represents the quantum research community that uses Rigetti for rapid prototyping but IBM/IonQ for production-grade experiments.

COMPOSITE / ILLUSTRATIVE — drawn from quantum computing academic community patterns

FINDING F-5-B: Portrait 2: The Fab-1 Process Engineer (COMPOSITE)

A materials scientist working in Rigetti's Fab-1 Fremont facility develops new Josephson junction geometries for higher coherence qubits. Her team iterates process recipes weekly — a speed advantage over companies that outsource fab. 'We can test a new qubit design in 3 weeks from concept to characterisation. External foundries would take 3 months.' She embodies the Fab-1 competitive advantage — continuous in-house process improvement at quantum speed.

COMPOSITE / ILLUSTRATIVE — drawn from superconducting qubit fabrication community patterns

FINDING F-5-C: Portrait 3: The Startup Using QCS for Optimisation (COMPOSITE)

A 28-year-old founder of a quantum chemistry startup uses Rigetti QCS for proof-of-concept simulations of electrolyte decomposition in EV batteries. He chose Rigetti because QCS offers the most developer-friendly API and the lowest cost per QPU hour for NISQ-era circuits. His concern: 'If IonQ's trapped-ion achieves fault tolerance first, I'll switch. Rigetti needs to keep pace.' He represents the startup ecosystem that bootstraps on QCS's accessibility.

COMPOSITE / ILLUSTRATIVE — drawn from quantum startup ecosystem patterns

PART VI — ADVERSARIAL COLLABORATION (5 HYPOTHESES)

HH1: HH1: IBM/Google make Rigetti irrelevant with scale superiority

EVIDENCE FOR:

IBM's 433-qubit Osprey and Google's Willow outperform Rigetti Aspen at every benchmark; IBM's 10× qubit scale advantage makes Rigetti hardware obsolete for serious research; enterprise customers choose IBM or Google over Rigetti for production

EVIDENCE AGAINST:

Rigetti's Fab-1 enables faster process iteration than IBM's internal roadmap; QCS's direct access and developer ecosystem provides niche stickiness; NISQ-era niche applications don't require IBM-scale hardware

QUANTITATIVE TEST

IBM's qubit advantage is decisive for >50 qubit circuits; P(Rigetti loses >50% enterprise accounts to IBM by 2028)=30%

VERDICT: BEAR — IBM scale threat is the most credible existential risk

Implication: Rigetti must differentiate on Fab-1 process innovation or be commoditised

HH2: HH2: Rigetti's Fab-1 achieves coherence times that leapfrog IBM**EVIDENCE FOR:**

Full in-house process control enables material science breakthroughs unavailable to outsourced fabs; new qubit geometries (fluxonium, Andreev) improve T1/T2; Rigetti achieves 99.9% two-qubit fidelity before IBM's next generation

EVIDENCE AGAINST:

IBM has 10× more engineers and 100× more capital; Rigetti's Fab-1 cannot outspend IBM's research division; scale of IBM's quantum network effects dominate

QUANTITATIVE TEST

Fidelity breakthroughs are possible on small-team focused effort; P(Rigetti achieves fidelity parity with IBM's top system by 2027)=20%

VERDICT: BULL — if coherence breakthrough, Rigetti re-rates significantly

Implication: Monitor Rigetti's quarterly qubit benchmarks vs IBM Quantum Volume

HH3: HH3: Rigetti runs out of cash before achieving quantum advantage**EVIDENCE FOR:**

Net income -\$233.6M TTM with limited revenue; Fab-1 maintenance has high fixed costs; government contracts insufficient to cover burn; equity raise at severe dilution

EVIDENCE AGAINST:

DARPA contract + QCS commercial revenue + government partnerships provide runway; Rigetti has demonstrated cash management discipline post-SPAC period

QUANTITATIVE TEST

P(cash crisis requiring rescue financing by 2027)=35%

VERDICT: BEAR — dilution or restructuring risk is material

Implication: Monitor quarterly cash balance and operating cash flow trend

HH4: HH4: Trapped-ion wins market, superconducting becomes legacy technology

EVIDENCE FOR:

IonQ's fidelity advantage + all-to-all connectivity makes superconducting obsolete for NISQ-era computation; quantum advantage demonstrated on trapped-ion first

EVIDENCE AGAINST:

Superconducting scales to 1,000+ qubits more naturally than trapped-ion; speed advantage (nanosecond gates) is decisive for error correction cycles; IBM and Google's enormous investment validates superconducting long-term

QUANTITATIVE TEST

P(trapped-ion replaces superconducting as dominant modality by 2030)=10%

VERDICT: NEUTRAL — modality war not decided; both paths viable at current state

Implication: Rigetti's technology bet on superconducting is structurally correct long-term

HH5: HH5: Strategic acquisition of Rigetti by AWS or defence contractor

EVIDENCE FOR:

Rigetti's Fab-1 is a strategic US asset for quantum sovereignty; AWS could acquire QCS and Fab-1 to vertical-integrate quantum hardware; Lockheed or Northrop could acquire for classified quantum computing

EVIDENCE AGAINST:

Antitrust risk for AWS acquisition; defence acquisitions typically at book value; Rigetti management historically resistant to acquisition approach

QUANTITATIVE TEST

P(acquisition at >50% premium to current price by 2028)=12%

VERDICT: BULL scenario — Fab-1 strategic asset justifies premium

Implication: Monitor government contract pipeline for strategic interest signals

PART VII — SUPERFORECASTING (10 SCENARIOS)

SCENARIO 1: Fab-1 process breakthrough — 99.9% fidelity achieved (P = 18%)

Rigetti's in-house fab achieves coherence improvement enabling fault-tolerant circuits; triggers re-rating to \$30+ range; DARPA validates and expands contract

Driver: Quarterly qubit benchmark disclosures; Nature/PRX Quantum publications

Falsifier: IBM Quantum Volume data shows Rigetti falling further behind

SCENARIO 2: QCS commercial revenue reaches \$20M ARR (P = 30%%)

QCS subscription growth from developer community; 3-5 new enterprise contracts in optimisation/simulation; AWS Braket usage fees generate royalty income

Driver: Quarterly revenue and customer count disclosures

Falsifier: QCS revenue below \$10M for two consecutive quarters

SCENARIO 3: Base case: survival — revenue grows, losses narrow slowly (P = 32%%)

Rigetti maintains DARPA contract, grows QCS to \$15M ARR; dilutive equity raise in 2027 maintains runway; stock range \$12-22; no decisive technical advantage

Driver: DARPA contract renewal; QCS growth rate

Falsifier: Revenue miss + dilution triggers 40%+ drawdown

SCENARIO 4: Quantum winter — stock falls below 52w low (\$12.53) (P = 20%%)

Broad quantum sentiment collapse; Rigetti revenue disappoints; equity raise at \$10-12; 52w low retested

Driver: Two consecutive revenue misses; Fab-1 upgrade delayed

Falsifier: Government contract announcement restores confidence

SCENARIO 5: IBM patents or regulatory action restricts Fab-1 (P = 5%%)

IBM IP dispute over superconducting qubit architecture; regulatory hurdle on Fab-1 operations (environmental); operational disruption to core manufacturing capability

Driver: Patent litigation filings; EPA regulatory actions

Falsifier: Rigetti wins IP dispute; Fab-1 receives clean environmental audit

SCENARIO 6: DoD classified quantum computing contract (\$50M+) (P = 15%%)

US military awards Rigetti classified quantum computing contract; US-made Fab-1 chips satisfy supply chain security requirements; government revenue exceeds commercial for first time

Driver: DoD budget cycle; classified programme announcements

Falsifier: Contract awarded to IBM or trapped-ion competitor

SCENARIO 7: Rigetti-IonQ technical partnership (P = 8%%)

Rigetti and IonQ form hybrid quantum architecture partnership; superconducting speed + trapped-ion fidelity in hybrid stack; joint DARPA/NSF proposal wins government funding

Driver: Industry partnership announcements; joint paper submissions

Falsifier: Companies remain direct competitors without collaboration

SCENARIO 8: Rigetti's 1000-qubit system (Lyra) delivers commercial advantage (P = 20%%)

Rigetti's roadmap 1000-qubit Lyra system launches commercially; first system to demonstrate practical quantum advantage at scale; IonQ partnership or competition intensifies

Driver: Rigetti engineering blog; investor day roadmap disclosure

Falsifier: Lyra delayed past 2028 or underperforms on benchmarks

SCENARIO 9: European government quantum contract (UK/Germany/France) (P = 12%%)

UK NQTP or German Quantum Initiative selects Rigetti QCS; European sovereign quantum cloud uses Rigetti hardware; opens \$15M+ ARR non-US government channel

Driver: European quantum programme tender announcements

Falsifier: Europe selects IQM (Finland) or IonQ for sovereign quantum

SCENARIO 10: Bull case: \$300M+ market cap acquisition + Fab-1 spinout (P = 6%%)

Private equity or defence contractor acquires Rigetti; Fab-1 spun out as independent quantum foundry; QCS platform sold to AWS; aggregate transaction value \$2-3B

Driver: Board governance change; activist investor involvement

Falsifier: No acquisition interest; Rigetti remains independent and loss-making

PART VIII — SELDON VARIABLE DEEP DIVE

Seldon Variable	Rigetti Impact	Direction	Magnitude
K (Knowledge Capital)	Full-stack superconducting accelerates quantum compute K-curve	↑ High potential	K++ at fault tolerance
Ω (Existential Risk)	Superconducting QC can run Shor's algorithm at scale — crypto threat	↑ Elevates Ω	Ω++ if scale achieved
I (Institutions)	Fab-1 sovereign manufacturing validates US	↑ Positive	I+ moderate

	quantum industrial base		
ξ (Distribution)	QCS accessible via AWS — marginally better quantum distribution	→ Neutral	ξ neutral

PART IX — RISK REGISTER

Risk	Probability	Impact	Mitigation
IBM/Google obsolescence risk	HIGH (30%)	CRITICAL	Fab-1 process differentiation; fidelity roadmap
Cash burn / dilution	HIGH (35%)	HIGH	Government contract pipeline; QCS ARR growth
Technology bet failure (superconducting loses)	MEDIUM (10%)	CRITICAL	Modality war unsettled; long-duration risk
Fab-1 operational disruption	LOW (10%)	HIGH	Backup supplier relationships; insurance
Talent risk — quantum engineers are scarce	MEDIUM (25%)	HIGH	Equity compensation; university partnerships

CONCLUSION — 6-METHOD SYNTHESIS VERDICT

CONVICTION VERDICT: HIGH RISK SPECULATIVE — FAB-1 IS THE MOAT, DILUTION IS THE RISK

Rigetti at \$18.81 (mkt cap \$6.278B) is the highest-risk / highest-reward pure-play superconducting quantum bet among listed companies. The bull case is Fab-1-driven fidelity breakthrough + government mega-contract. The bear case is IBM/Google make Rigetti irrelevant while cash burns. Seldon alignment: K-variable potential is high; Ω-risk elevated (Shor's algorithm threat). HUMAN_ONLY_MANUAL — CIO decides position. All figures: moomoo OpenD [2026-08-13]. RAG Corpus: EDGAR 139,756ch · WSJ 335,496ch · FT 109,507ch · NIKKEI 397,454ch · FA 148,327ch · ST_PHD 60,492ch.